SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

KAVIYAASRI M
192312614RB

CO5 - AT1 - LEVEL-3

Level Exam | </> C | 02 Sep 2026, 02:45 PM

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Result Available
Your score: **80 / 100**

QUESTIONS

LEVEL 3 — HARD

- Q1 — function
- Q2 — functionsandpointers

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Level 3 — Hard **power (H)** — function

```
Write a function double power(double base, int exponent) that calculates the power of a given base raised to a given exponent.
        Sample Input:
Enter the base: 2
Enter the exponent: 3

Sample Output:
2 raised to the power 3 is 8
Number of consonants: 7
```

Your Code

```c
#include <stdio.h>
#include <ctype.h>
double power(double base, int exponent) {
    double result = 1.0;
    for (int i = 0; i < (exponent >= 0 ? exponent : -expon
        result *= base;
    }
    if (exponent < 0) result = 1.0 / result;
    return result;
}
int countConsonants(const char *s) {
    int count = 0;
```

Program Input (stdin)

Enter input if required...

Output

```
Enter the base:
Enter the exponent:
2 raised to the power 3 is 8
Number of consonants: 7
```

Your Code

```
#include <stdio.h>
#include <ctype.h>
double power(double base, int exponent) {
    double result = 1.0;
    for (int i = 0; i < (exponent >= 0 ? exponent : -expon
        result *= base;
    }
    if (exponent < 0) result = 1.0 / result;
    return result;
}
int countConsonants(const char *s) {
    int count = 0;
    for (int i = 0; s[i] != '\0'; i++) {
        char c = tolower(s[i]);
        if (c >= 'a' && c <= 'z') {
            if (c != 'a' && c != 'e' && c != 'i' && c != '
                count++;
        }
    }
    return count;
}
int main() {
    double base;
    int exponent;
    printf("Enter the base: ");
    scanf("%lf", &base);
    printf("\nEnter the exponent: ");
    scanf("%d", &exponent);
    double ans = power(base, exponent);
    printf("\n%.0f raised to the power %d is %.0f\n", base
```

Program Input (stdin)

Enter input if required...

Output

```
Enter the base:
Enter the exponent:
0 raised to the power 0 is 1
Number of consonants: 7
```

Run Save

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Your Code

```
#include <stdio.h>
int isPalindrome(char *str) {
    char *end = str;
    char *start = str;
    while (*end != '\0') {
        end++;
    }
    end--;
    while (start < end) {
        if (*start != *end) {
            return 0;
        }
        start++;
        end--;
    }
    return 1;
}
int main() {
    char str[100];
    printf("Enter a string: ");
    scanf("%99s", str);
    if (isPalindrome(str)) {
        printf("\n'%s' is a palindrome\n", str);
    } else {
        printf("\n'%s' is not a palindrome\n", str);
    }
    return 0;
}
```

Program Input (stdin)

Enter input if required...

Output

```
Enter a string:
'racecar' is a palindrome
```

Run Save